```
population_size = 500
2
mutation_rate = 0.01
pop_fit = []
pop = spawn_population(N=population_size,size=407)
3
for i in range(num_generations):
    pop, avg_fit = evaluate_population(pop)
4
    pop_fit.append(avg_fit)
    pop = next_generation(pop,
mut_rate=mutation_rate,tournament_size=0.2) 5
```

- ***1*** The number of generations to evolve
- ***2*** The number of individuals in each generation
- ***3*** Initializes a population
- ***4*** Evaluates the fitness of each agent in the population
- ***5*** Populates the next generation

The first generation begins with a population of random parameter vectors, but by chance some of these will be better than the others, and we preferentially select these to mate and produce offspring for the next generation. To maintain genetic diversity, we allow each individual to be mutated slightly. This process repeats until we have individuals who are exceptionally good at playing CartPole. You can see in figure 6.9 that the score steadily increases each generation of evolution.

Figure 6.9. The average score of the population over generations in a genetic algorithm used to train agents to play CartPole.